

Navigating the waters of nixtamalization: Sustainable solutions for maize-processing wastewater treatment

Author links open overlay panelEdgardo I. Valenzuela ^a, [Janet A. Gutiérrez-Uribe](#) ^{a, b}, Mariana Franco-Morgado ^{a, b}, Pabel Cervantes-Avilés ^a

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Highlights

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Physicochemical traits of nejayote from distinct sources importantly differ.

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Current nejayote treatment methods have drawbacks due to its high strength.

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Highly alkaline pH and Ca²⁺ concentrations hinder all nejayote treatments.

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Combining biological and physicochemical treatments may enhance efficacy.

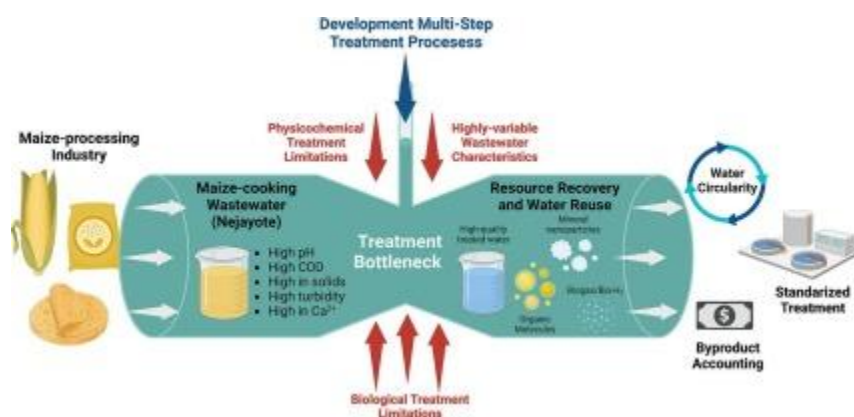
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Mineral resources can be recovered from nejayote liquid and solid phases.

Abstract

Maize-processing wastewater, also known as *nejayote*, does represent a widespread residue originating from both small- and large-scale factories that produce maize-based products using the alkaline maize-cooking process ([nixtamalization](#)). *Nejayote* is a high-strength wastewater containing significant concentrations of soluble and insoluble organic and [inorganic compounds](#) resulting from the disintegration of maize, as well as from the substantial quantity of [lime](#) (Ca(OH)₂) used in the process. In order to make [nixtamalization](#) more sustainable in terms of water usage and to mitigate the health and [environmental issues](#) related to nejayote discharges into environmental matrices and public [sewage systems](#), appropriate and effective treatment processes must be applied

either before effluent disposal or for [water reuse](#) purposes. With this problematic as the central topic, we conducted a comprehensive review of relevant literature addressing this issue spanning from the mid-1980s to the present day. This review covers three primary aspects: i) the extensive variability observed in the physicochemical composition of maize-processing wastewater, ii) the various biological and physicochemical methods developed for its treatment, and iii) the potential for organic and mineral resource recovery from this waste. Although initial efforts to treat nejayote were left behind for decades, recent years have witnessed a resurgence of research interest in these research topics mainly underpinned by the urgency to conserve [water resources](#). Based on the comprehensive evaluation of the existing literature, we identified the existing limitations on nejayote treatment and identified prospects for developing robust and technically feasible treatment possibilities. Within this review, we propose three main approaches for [wastewater treatment](#) and [water reuse](#): physicochemical-based technologies, bioprocess in tandem with [membrane technology](#), and low-cost bioprocesses coupled to physicochemical methods.



Introduction

corn kernels (Santiago-Ramos et al., 2018). *Nejayote*, also known as *maize processing wastewater* or *corn/lime cooking liquor*, is the byproduct of nixtamalization and consists of a highly turbid, yellow-colored, starchy liquid that remains after the boiling and steeping process. Due to its unique and extremely variable physicochemical characteristics, including high pH and considerable content of suspended and dissolved organic compounds, nejayote represents an environmental threat, as it is usually discharged into water bodies and public sewers in large quantities (Argun and Argun, 2018; Castro-Muñoz et al., 2019). For instance, Rojas-García et al. (2012) have estimated that in Mexico around 14.4 million m³ of nejayote is produced annually. This considers that a maize-processing factory with a 600 ton maize/day capacity generates between 1500 and 2000 m³ of nejayote (Salmerón-Alcocer et al., 2003; Rojas-García et al., 2012). Due to its strong maize-based dietary traditions, Mexico is the main country in which nejayote represents a significant issue. However, Central and South American countries, such as Honduras, El Salvador, Guatemala, Colombia, and Venezuela, are also significant consumers of nixtamalized maize-based products, potentially leading to the generation of significant amounts of maize- processing wastewater needing treatment (Erenstein et al., 2022). Moreover, apart from major food corporations producing savory snacks based on nixtamalized maize or dry masa flour (DMF) (i.e., tortilla-based chips), the increasing popularity of Mexican and Latin American cuisine have led to the establishment of DMF plants involving the nixtamalization process in various countries, including Russia, Korea, China, Australia, the US, and several European countries (Serna-Saldivar and Chuck-Hernandez, 2019).

The environmental impacts of the maize-processing industry have just been recently documented (Guzmán-Soria et al., 2019; Quintero-Herrera et al., 2023). Considering the significant amounts of nejayote produced, its inappropriate disposal and the lack of proper treatment, discharges of nejayote may lead to important ecological disturbances such as eutrophication, degradation of aquatic ecosystems, and the proliferation of pests and insects. These disturbances can ultimately result in the spread of diseases and pose public health issues (Bahri et al., 2009; Khan and Mohammad, 2014). Additionally, due to its high content of solids and dissolved calcium, untreated nejayote disposal may have detrimental effects on the piping and facilities of industrial and public sewage systems. These effects can include total or partial piping blockage/reduced capacity, formation of solid deposits or scale (due to calcium-bearing mineral accumulation) and corrosion (Hammes et al., 2003; Sanciolo et al., 2008). These issues can lead to an increased maintenance and cleaning requirements and reduced lifespan of equipment, resulting in increased operational costs and disruptions to the sewage systems. There are no documented records about scaling effects specifically for nejayote discharges, although these effects have been extensively recorded when industrial wastewaters of similar characteristics are disposed without

treatment. Considering that nixtamal factories are widespread in countries where maize-based products are commercialized, greater efforts to make this type of industry cleaner and sustainable in face of the ongoing climate and water crises must be made.

For these reasons, the need for effective treatment of nejayote before disposal in water bodies has been a significant concern for decades, with research efforts dating back to the late 1980s (Civit et al., 1984; González-Martínez, 1984). However, despite some initial progress, these efforts were left behind for years (Fig. 1). It was not until the late 2000s that interest in the treatment of nejayote emerged considerably, prompted by the recognition of the environmental and public threat that this effluent represents, along with the increasing need of designing effective wastewater treatment processes to alleviate the ongoing water crisis and increase its reuse (Argun and Argun, 2018; Castro-Muñoz et al., 2019). Several options have been explored to tackle the complexity of nejayote for treatment or resource recovery. Such efforts include physicochemical and biological treatment to diminish its pollutant concentration (López-Pacheco et al., 2019; Valderrama-Bravo et al., 2022), biological processes for bioenergy production (García-Depraect et al., 2017; Del Angel-Acosta et al., 2021), as well as high-end filtration processes for the recovery of organic molecules of biotechnological interest (OMBI) (i.e., antioxidants, food additives, etc.) (Castro-Muñoz et al., 2019; Contreras-Jáquez et al., 2022a).

Despite efforts to address nejayote treatment, there are currently no standardized techniques applied in maize-processing factories, with or without options for products revalorization or water reuse. This might be mainly due to several drawbacks originated by the complex and widely variable physicochemical composition of nejayote. Such disadvantages result in low-efficiency processes requiring pretreatments (i.e., enzymatic) (García-Zamora et al., 2015), high energy inputs (e.g., for membrane filtrations or reverse osmosis) (Contreras-Jáquez et al., 2022a; Valderrama-Bravo et al., 2022), and often the dilution with other types of effluents, even with clean and tap-water (García-Depraect et al., 2017; López-Pacheco et al., 2019; Del Angel-Acosta et al., 2021). Additionally, there are no proposals for treatment trains that combine the benefits of biological and physicochemical approaches for byproducts recovery and water reuse, although this could effectively address the disadvantages of each individual treatment technique. Moreover, an additional aspect to take in consideration is the size and location of maize-processing facilities. For instance, in Mexico, maize-cooking factories producing nixtamal, DMF, or tortillas, can vary widely in both size and location. For instance, larger-scale maize-processing facilities are usually situated within industrial parks in major cities. Contrary case, numerous smaller factories are commonly found in middle- and small-sized cities as well as rural towns. Also, some these towns can be located in remote and geographically isolated areas. In the context of wastewater treatment, implementing effective processes

for nejayote treatment does imply unique challenges such as limited access to infrastructure, technical expertise, and financial resources. Therefore, choosing a methodology to effectively treat nejayote must take these variables into consideration.

For these reasons, the existing literature about nejayote physicochemical characterization, treatment and valorization strategies was reviewed in this paper. For the purpose of this review, we thoroughly revised the existing body of knowledge on “nejayote” and “maize-processing wastewater”, encompassing scientific papers published from the mid-1980s to the present day. The identified literature was thoroughly reviewed and categorized into six main categories: *food science*, *biological treatment*, *physicochemical treatment*, *organic molecules of biotechnological interest recovery*, *biological sciences*, and *materials sciences* (as depicted in Fig. 1 and Supplementary Materials). All publications containing complete or partial nejayote physicochemical characterizations were considered for making an analysis on the highly variable composition that this residue presents. Furthermore, critically assessed those papers focused in biological and physicochemical treatment, along with those addressing the recovery of organic molecules. This thorough examination aimed to identify opportunities, limitations, research gaps, and new approaches as well as directions for nejayote treatment. Finally, driven by the goals of water reuse and resource recovery, we provide novel strategies that combine several combinations of physicochemical and biological approaches. These strategies have the objective of enhancing the sustainability of the nixtamalization process by mitigating hydric stress in times of water crisis.

Physicochemical characteristics of nejayote

There are numerous studies in the literature reporting the main components and physicochemical parameters of nejayote. These parameters are associated to negative impacts on the aquatic environment and difficulties related to their removal. A compendium of the most significant parameters from the wastewater perspective is summarized in Fig. 2 (raw data are provided in Table S1 in Supplemental Material).

The main parameters to be considered for treatment of nejayote include dissolved Ca^{2+} ,

Aerobic systems

The first efforts for biological treatment of nejayote under aerobic conditions emerged in the middle of the 1980s and regained interest in the beginning of the 2000s (Fig. 1). Bioreactor configurations employed for this purpose included bench- and pilot-scale aerated tanks (González-Martínez, 1984), rotating biological contactors (RBC) (de Brenes and de Bazúa, 1987; Pedroza-Islas and Durán de Bazúa, 1990; Luna-Pabello et al., 1996), submerged fixed-film cascade bioreactors (SFFCR), as well as

Recovery of organic molecules of biotechnological interest

Presence of biomolecules in nejayote with high commercial and environmental interests have been reported for more than three decades (Table 3). Detection of essential amino acids such as lysine, threonine, isoleucine, leucine, valine, methionine and phenylalanine, and non-essential amino acids including histidine, aspartic acid, glycine, glutamic acid, arginine, proline, were first reported by de Brenes and de Bazúa (1987). However, limited work has been done to recover these metabolites.

Limitations and potential strategies for water reuse

With water becoming a scarce and valuable resource, industries, including food-manufacturing businesses such as maize-processing factories, must seek strategies to incorporate water reuse within their processes (e.g., within the nixtamalization process) (Fig. 3). By doing so, industries can diminish its dependance on freshwater sources by treating and reusing water generated from their process stages (e.g., nejayote). This water-circularity approach also represents a cost-saving opportunity by

Conclusions and perspectives

The challenge in treating nejayote comes from its unique characteristics such as its highly alkaline pH, elevated levels of Ca^{2+} , COD, and solids content, but also from its content of organic and inorganic pollutants which vary significantly from source to source. Despite efforts to obtain OMBIs from nejayote through separation techniques, as well as for eliminating pollutants via biological methods, high operational costs and low treatment efficiencies limit the widespread implementation of

CRedit authorship contribution statement

Edgardo I. Valenzuela: Writing – Original Draft, Visualization, Conceptualization, Methodology, Investigation, Project administration, **Janet A. Gutiérrez-Urbe:** Writing – Review & Editing, Investigation, **Mariana Franco-Morgado:** Writing – Review & Editing, Investigation, **Pabel Cervantes-Aviles:** Writing - Review & Editing, Conceptualization, Investigation, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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